APM1110: Probability and Probability Distributions

Formative Assessment 8 — Complete Solutions

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Problem 1 (6 points)

Setup. Total balls: $10 + 8 + 7 = 25$. Let C be the chosen color, with $P(C = \text{red}) = P(C = \text{black}) = P(C = \text{green}) = \frac{1}{3}$. Then 4 balls are drawn without replacement, and X counts how many match C.

Combinations needed (using $\binom{n}{4} = \frac{n(n-1)(n-2)(n-3)}{4!}$, $4! = 24$):

$$\begin{aligned}
\binom{25}{4} &= \frac{25 \cdot 24 \cdot 23 \cdot 22}{24} = \frac{303600}{24} = 12650, \\
\binom{15}{4} &= \frac{15 \cdot 14 \cdot 13 \cdot 12}{24} = \frac{32760}{24} = 1365, \\
\binom{17}{4} &= \frac{17 \cdot 16 \cdot 15 \cdot 14}{24} = \frac{57120}{24} = 2380, \\
\binom{18}{4} &= \frac{18 \cdot 17 \cdot 16 \cdot 15}{24} = \frac{73440}{24} = 3060.
\end{aligned}$$

(a) $P(X = 0)$. By the law of total probability, conditioning on the chosen color:

$$P(X = 0) = \sum_{c} P(X = 0 \mid C = c)\, P(C = c).$$

$X = 0$ means all 4 balls come from the balls *not* of the chosen color:

- Red chosen: $25 - 10 = 15$ non-red balls, so $P(X = 0 \mid \text{red}) = \binom{15}{4}/\binom{25}{4} = 1365/12650$.
- Black chosen: $25 - 8 = 17$ non-black balls, so $P(X = 0 \mid \text{black}) = 2380/12650$.
- Green chosen: $25 - 7 = 18$ non-green balls, so $P(X = 0 \mid \text{green}) = 3060/12650$.

$$P(X = 0) = \frac{1}{3} \cdot \frac{1365 + 2380 + 3060}{12650} = \frac{1}{3} \cdot \frac{6805}{12650} = \frac{6805}{37950} = 0.17931\ldots$$

$$\boxed{P(X = 0) \approx 0.1793}$$

(b) $P(X_i = 1)$. The 4 balls are drawn at random, so the ith ball selected is equally likely to be any of the 25 balls (each ball has probability $1/25$ of being in position i). Given the chosen color, the probability that ball i matches it is (number of balls of that color)/25:

$$P(X_i = 1) = \tfrac{1}{3} \cdot \tfrac{10}{25} + \tfrac{1}{3} \cdot \tfrac{8}{25} + \tfrac{1}{3} \cdot \tfrac{7}{25} = \tfrac{1}{3} \cdot \frac{10 + 8 + 7}{25} = \tfrac{1}{3} \cdot \frac{25}{25} = \tfrac{1}{3}.$$

This holds for each $i = 1, 2, 3, 4$:

$$\boxed{P(X_i = 1) = \tfrac{1}{3} \approx 0.3333, \quad i = 1, 2, 3, 4}$$

(c) $E[X]$. Since X_i indicates whether the ith ball matches the chosen color, $X = X_1+X_2+X_3+X_4$. For an indicator, $E[X_i] = 1 \cdot P(X_i = 1) + 0 \cdot P(X_i = 0) = \frac{1}{3}$. By linearity of expectation (no independence needed):

$$E[X] = E[X_1] + E[X_2] + E[X_3] + E[X_4] = 4 \cdot \tfrac{1}{3} = \tfrac{4}{3}.$$

$$\boxed{E[X] = \tfrac{4}{3} \approx 1.3333}$$

Problem 2 (8 points)

$P(X = n) = \dfrac{4}{n(n+1)(n+2)}$, $n \geq 1$.

Verifying the hints.

$$\frac{1}{2}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right] = \frac{1}{2} \cdot \frac{(n+2)-n}{n(n+1)(n+2)} = \frac{1}{2} \cdot \frac{2}{n(n+1)(n+2)} = \frac{1}{n(n+1)(n+2)},$$

$$\frac{k}{n(n+k)} = \frac{(n+k)-n}{n(n+k)} = \frac{1}{n} - \frac{1}{n+k}.$$

(a) Show $\sum_{n=1}^{\infty} P(X = n) = 1$. All terms are positive. Using the first hint,

$$\sum_{n=1}^{\infty} \frac{4}{n(n+1)(n+2)} = 2\sum_{n=1}^{\infty}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right].$$

Let $a_n = \frac{1}{n(n+1)}$; the bracket is $a_n - a_{n+1}$. The partial sum telescopes:

$$S_N = 2\sum_{n=1}^{N}(a_n - a_{n+1}) = 2\,(a_1 - a_{N+1}) = 2\left[\frac{1}{2} - \frac{1}{(N+1)(N+2)}\right].$$

As $N \to \infty$, $\frac{1}{(N+1)(N+2)} \to 0$, so

$$\sum_{n=1}^{\infty} P(X = n) = 2 \cdot \tfrac{1}{2} = 1. \qquad \blacksquare$$

Since each $P(X = n) > 0$ and the total is 1, this is a valid PMF.

(b) Show $E[X] = 2$.

$$E[X] = \sum_{n=1}^{\infty} n \cdot \frac{4}{n(n+1)(n+2)} = 4\sum_{n=1}^{\infty} \frac{1}{(n+1)(n+2)}.$$

Using the second hint with $k = 1$ (and $n \to n+1$): $\frac{1}{(n+1)(n+2)} = \frac{1}{n+1} - \frac{1}{n+2}$. Then

$$\sum_{n=1}^{N}\left(\frac{1}{n+1} - \frac{1}{n+2}\right) = \frac{1}{2} - \frac{1}{N+2} \xrightarrow{N\to\infty} \frac{1}{2}.$$

Therefore

$$E[X] = 4 \cdot \tfrac{1}{2} = 2. \qquad \blacksquare$$

(c) Show $E[X^2] = \infty$.

$$E[X^2] = \sum_{n=1}^{\infty} n^2 \cdot \frac{4}{n(n+1)(n+2)} = 4\sum_{n=1}^{\infty} \frac{n}{(n+1)(n+2)}.$$

For $n \geq 1$: $n+1 \leq 2n$ and $n+2 \leq 3n$, so $(n+1)(n+2) \leq 6n^2$ and

$$\frac{n}{(n+1)(n+2)} \;\geq\; \frac{n}{6n^2} = \frac{1}{6n}.$$

Hence

$$E[X^2] \;\geq\; \frac{4}{6}\sum_{n=1}^{\infty} \frac{1}{n} = \infty,$$

because the harmonic series diverges. (Check: $\frac{n}{(n+1)(n+2)} = \frac{2}{n+2} - \frac{1}{n+1}$, whose partial sums grow like $\ln N$.) By comparison, $E[X^2] = \infty$. $\blacksquare$

Problem 3 (4 points)

$f(x) = a + bx^2$ for $0 \leq x \leq 1$, and $E[X] = \frac{3}{5}$.

Equation 1 (total probability is 1):

$$\int_0^1 (a + bx^2)\,dx = \Big[ax + \frac{b}{3}x^3\Big]_0^1 = a + \frac{b}{3} = 1.$$

Equation 2 (mean is $3/5$):

$$E[X] = \int_0^1 x(a + bx^2)\,dx = \int_0^1 (ax + bx^3)\,dx = \Big[\frac{a}{2}x^2 + \frac{b}{4}x^4\Big]_0^1 = \frac{a}{2} + \frac{b}{4} = \frac{3}{5}.$$

Solve. From Equation 1, $a = 1 - \frac{b}{3}$. Substitute into Equation 2:

$$\frac{1}{2}\left(1 - \frac{b}{3}\right) + \frac{b}{4} = \frac{3}{5} \;\Rightarrow\; \frac{1}{2} - \frac{b}{6} + \frac{b}{4} = \frac{3}{5} \;\Rightarrow\; \frac{1}{2} + \frac{b}{12} = \frac{3}{5}.$$

(Using $-\frac{b}{6} + \frac{b}{4} = \frac{-2b+3b}{12} = \frac{b}{12}$.) Then

$$\frac{b}{12} = \frac{3}{5} - \frac{1}{2} = \frac{6-5}{10} = \frac{1}{10} \;\Rightarrow\; b = \frac{12}{10} = \frac{6}{5}.$$

$$a = 1 - \frac{1}{3}\cdot\frac{6}{5} = 1 - \frac{2}{5} = \frac{3}{5}.$$

Check: $a + b/3 = 0.6 + 0.4 = 1$ and $a/2 + b/4 = 0.3 + 0.3 = 0.6$. Also $f(x) = 0.6 + 1.2x^2 \geq 0$ on $[0, 1]$.

$$\boxed{a = \tfrac{3}{5} = 0.6000, \qquad b = \tfrac{6}{5} = 1.2000}$$

Problem 4 (4 points)

$E[X] = \mu$, $\mathrm{Var}(X) = \sigma^2$, and $Y = \dfrac{X-\mu}{\sigma} = \dfrac{1}{\sigma}X - \dfrac{\mu}{\sigma}$, where μ, σ are constants.

Expected value. Using linearity, $E[aX+b] = aE[X] + b$:

$$E[Y] = \frac{1}{\sigma}E[X] - \frac{\mu}{\sigma} = \frac{\mu}{\sigma} - \frac{\mu}{\sigma} = 0.$$

Variance. Using $\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$ with $a = 1/\sigma$, $b = -\mu/\sigma$ (the constant b does not change variance):

$$\mathrm{Var}(Y) = \left(\frac{1}{\sigma}\right)^2 \mathrm{Var}(X) = \frac{1}{\sigma^2}\cdot\sigma^2 = 1.$$

$$\boxed{E[Y] = 0, \qquad \mathrm{Var}(Y) = 1}$$

Problem 5 (8 points)

$X \sim \mathrm{Poisson}(\lambda_1 = 2)$, $Y \sim \mathrm{Poisson}(\lambda_2 = 3)$, independent, $W = X + Y$.

(a) MGF of W. First, the MGF of a Poisson(λ) variable N, with $P(N=k) = e^{-\lambda}\lambda^k/k!$:

$$M_N(t) = \sum_{k=0}^{\infty} e^{tk}\frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty}\frac{(\lambda e^t)^k}{k!} = e^{-\lambda}e^{\lambda e^t} = e^{\lambda(e^t-1)}.$$

So $M_X(t) = e^{2(e^t-1)}$ and $M_Y(t) = e^{3(e^t-1)}$. By independence,

$$M_W(t) = E[e^{t(X+Y)}] = E[e^{tX}]\,E[e^{tY}] = M_X(t)M_Y(t) = e^{2(e^t-1)}e^{3(e^t-1)} = e^{(2+3)(e^t-1)}.$$

$$\boxed{M_W(t) = e^{5(e^t-1)}}$$

(b) Distribution of W. $M_W(t) = e^{5(e^t-1)}$ has exactly the Poisson MGF form with $\lambda = 5$. By the uniqueness property of MGFs (an MGF finite near 0 determines the distribution),

$$\boxed{W \sim \mathrm{Poisson}(\lambda = 5)}$$

(c) Moments. Differentiate $M_W(t) = e^{5(e^t-1)}$ using the chain rule:

$$M_W'(t) = e^{5(e^t-1)}\cdot 5e^t = 5e^t\,e^{5(e^t-1)}.$$

At $t = 0$ (note $e^{5(e^0-1)} = e^0 = 1$):

$$E[W] = M_W'(0) = 5\cdot 1\cdot 1 = 5.$$

Second derivative, by the product rule on $5e^t \cdot e^{5(e^t-1)}$:

$$\begin{aligned} M_W''(t) &= 5e^t\,e^{5(e^t-1)} + 5e^t\cdot\left(5e^t\,e^{5(e^t-1)}\right) \\ &= 5e^t\,e^{5(e^t-1)} + 25e^{2t}\,e^{5(e^t-1)}. \end{aligned}$$

At $t = 0$:

$$E[W^2] = M_W''(0) = 5 + 25 = 30.$$

Check: $\mathrm{Var}(W) = E[W^2] - (E[W])^2 = 30 - 25 = 5 = \lambda$, as expected for a Poisson variable.

$$\boxed{E[W] = 5, \qquad E[W^2] = 30}$$